

EA-0016 - AQELYN Engineering Archive

AQELYN - EA-0016 Engineering Archive

IS-016 - AQELYN Digital Forensics Engine

****Archive ID:**** EA-0016

****Implementation Specification:**** IS-016

****Component:**** AQELYN Digital Forensics Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0015

****Next Specification:**** IS-017 - AQELYN Threat Detection & Analytics Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0016 |

| Specification | IS-016 - AQELYN Digital Forensics Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0016.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-016 complete; EA-0016 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine
EA-0015	IS-015	AQELYN Security Operations (SOC) Engine
EA-0016	IS-016	AQELYN Digital Forensics Engine

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-016 Specification Identity

```
Specification ID: IS-016  
Name: AQELYN Digital Forensics Engine  
Engineering Archive Target: EA-0016  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-015 - AQELYN Security Operations Engine
```

3. Purpose

The AQELYN Digital Forensics Engine provides forensic acquisition, preservation, examination, analysis, and reporting capabilities across the AQELYN platform.

It ensures that evidence collected during security operations can be preserved with full integrity, chain of custody, and legal defensibility.

It answers:

- What happened?
- When did it happen?
- Who performed the action?
- Which systems were affected?
- What evidence supports the conclusion?
- Has the evidence been altered?
- Can the investigation be reproduced?
- Can the findings be presented for legal or regulatory review?

4. Mission

The engine shall provide:

- Digital evidence acquisition
- Evidence preservation
- Chain of custody
- Forensic imaging
- Artifact collection
- Timeline reconstruction
- Memory analysis
- Disk analysis
- Log analysis
- Evidence correlation
- Forensic reporting
- Court-ready documentation
- Evidence integrity verification

5. Scope

5.1 In Scope

- Digital forensic investigations
- Chain of custody management
- Artifact indexing
- Evidence hashing
- Evidence verification
- Timeline analysis
- Host forensics
- File system analysis
- Memory acquisition
- Registry analysis
- Browser artifact analysis
- Log correlation
- Forensic reporting
- Evidence export

5.2 Out of Scope

- Live malware analysis sandbox
- Reverse engineering framework
- Network packet capture engine
- Physical laboratory equipment
- Hardware write blockers
- Commercial forensic licensing

6. Dependencies

IS-016 depends on:

- IS-001 AQELYN Kernel
- IS-002 Universal Object Model

```
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
```

7. High-Level Architecture

```
AQELYN Digital Forensics Engine
■
■■■ Acquisition Manager
■■■ Chain of Custody Manager
■■■ Evidence Verification Service
■■■ Artifact Repository
■■■ Timeline Engine
■■■ Memory Analysis Engine
■■■ Disk Analysis Engine
■■■ Log Analysis Engine
■■■ Report Generator
■■■ Evidence Export Service
■■■ Knowledge Graph Connector
■■■ Evidence Engine Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-016-001 - Evidence Acquisition

The engine shall support acquisition of:

```
Disk images
Memory images
System logs
Audit logs
Registry data
Browser artifacts
File metadata
Security event artifacts
```

FR-016-002 - Chain of Custody

Every evidence item shall maintain:

```
Unique identifier
Collection timestamp
Collector identity
Evidence hash
Storage location
Transfer history
Verification history
```

FR-016-003 - Evidence Verification

The engine shall verify integrity using cryptographic hashing.

Supported capabilities:

- SHA-256
- SHA-512
- Hash comparison
- Integrity validation
- Verification logging

FR-016-004 - Timeline Reconstruction

The engine shall reconstruct chronological event timelines using collected artifacts.

FR-016-005 - Artifact Analysis

Support analysis of:

- Filesystem metadata
- Registry
- Browser history
- Event logs
- Memory structures
- Authentication events
- Security alerts

FR-016-006 - Reporting

Generate standardized forensic reports including:

- Evidence summary
- Chain of custody
- Timeline
- Findings
- Evidence references
- Integrity verification
- Analyst notes
- Recommendations

FR-016-007 - Event Publication

Publish forensic events:

- evidence.acquired
- evidence.verified
- timeline.generated
- analysis.completed
- report.generated

9. Non-Functional Requirements

The engine shall provide:

- Immutable evidence handling
- Cryptographic verification
- Full auditability
- Evidence traceability
- Scalable storage integration
- Repository stability
- Backward compatibility
- High availability

10. Core Forensic Workflow

- Evidence Acquired
- ↓
- Integrity Verified

```
↓
Chain of Custody Recorded
↓
Artifact Analysis
↓
Timeline Reconstruction
↓
Evidence Correlation
↓
Report Generated
↓
Evidence Archived
```

11. Internal Component Architecture

The AQELYN Digital Forensics Engine is implemented as a modular subsystem integrated with the AQELYN Kernel, Evidence Engine, Knowledge Graph, and Event Bus.

```
AQELYN Digital Forensics Engine
■
■■■ Acquisition Manager
■■■ Chain of Custody Manager
■■■ Evidence Verification Service
■■■ Artifact Repository
■■■ Timeline Engine
■■■ Memory Analysis Engine
■■■ Disk Analysis Engine
■■■ Log Analysis Engine
■■■ Registry Analysis Engine
■■■ Browser Artifact Engine
■■■ Report Generator
■■■ Evidence Export Service
■■■ Knowledge Graph Connector
■■■ Evidence Engine Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Acquisition Manager

Responsible for forensic acquisition.

Functions:

```
Evidence collection
Acquisition validation
Metadata recording
Evidence registration
Hash generation
```

12.2 Chain of Custody Manager

Maintains complete custody history.

Responsibilities:

```
Custody creation
Transfer logging
Ownership tracking
Access recording
Integrity verification
```

12.3 Evidence Verification Service

Performs cryptographic verification.

Supported operations:

- SHA-256 verification
- SHA-512 verification
- Integrity validation
- Evidence comparison

12.4 Artifact Repository

Stores indexed forensic artifacts.

Supported artifact types:

- Files
- Memory images
- Disk images
- Logs
- Registry
- Browser artifacts
- Metadata

12.5 Timeline Engine

Constructs chronological investigation timelines.

Sources include:

- Filesystem timestamps
- Audit logs
- Security events
- Authentication records
- Evidence metadata

12.6 Memory Analysis Engine

Supports forensic memory analysis.

Capabilities:

- Process enumeration
- Network connections
- Loaded modules
- Credential artifacts
- Memory anomalies

12.7 Disk Analysis Engine

Supports disk image analysis.

Capabilities:

- Filesystem analysis
- Deleted file recovery
- Partition inspection
- Metadata extraction
- Hash verification

12.8 Log Analysis Engine

Supports forensic log correlation.

Sources:

- Operating system logs
- Application logs
- Security logs
- Audit logs
- AQELYN events

12.9 Registry Analysis Engine

Supports registry artifact analysis.

Includes:

- Autoruns
- Installed software
- User activity
- Persistence indicators
- Configuration changes

12.10 Browser Artifact Engine

Analyzes browser artifacts.

Supports:

- History
- Downloads
- Cookies
- Cache
- Bookmarks
- Session data

12.11 Report Generator

Produces standardized forensic reports.

Outputs:

- PDF
- HTML
- Markdown
- JSON

12.12 Evidence Export Service

Exports forensic evidence.

Supported formats:

- ZIP
- JSON
- CSV
- Evidence package

13. Universal Object Model Extensions

13.1 EvidenceArtifact

```
EvidenceArtifact:  
  artifact_id  
  evidence_id  
  artifact_type
```


source
hash

13.2 ChainOfCustody

ChainOfCustody:
custody_id
evidence
collector
timestamp
transfers

13.3 Timeline

Timeline:
timeline_id
events
evidence

13.4 ForensicReport

ForensicReport:
report_id
investigation
findings
evidence

14. Knowledge Graph Integration

Relationships:

Evidence
↓
contains
↓
Artifact

Artifact
↓
belongs_to
↓
Timeline

Timeline
↓
supports
↓
Investigation

Investigation
↓
supports
↓
Incident

Evidence
↓
verified_by
↓
Hash

15. Event Bus Integration

15.1 Evidence Events

```
evidence.acquired  
evidence.verified  
evidence.exported
```

15.2 Analysis Events

```
analysis.started  
analysis.completed  
timeline.generated
```

15.3 Reporting Events

```
report.generated  
report.exported
```

16. Evidence Engine Integration

Consumes and extends evidence managed by IS-004.

Provides:

```
Artifact metadata  
Hash validation  
Chain of custody  
Evidence references
```

17. Knowledge Graph Integration Details

Publishes forensic relationships for:

```
Artifacts  
Evidence  
Investigations  
Incidents  
Threats  
Assets  
Identities
```

18. Security Operations Integration

Supports IS-015 by providing:

```
Incident evidence  
Investigation artifacts  
Timeline analysis  
Forensic reports  
Evidence verification
```

19. Risk Intelligence Integration

Provides:

```
Evidence supporting risk  
Incident impact validation  
Risk investigation artifacts
```

20. Compliance Integration

Supports:

- Audit evidence
- Compliance investigations
- Control validation
- Regulatory reporting

21. Policy Integration

Policies govern:

- Evidence retention
- Evidence access
- Custody rules
- Export authorization
- Destruction policy

22. Public APIs

22.1 Evidence API

- GET /forensics/evidence
- POST /forensics/evidence
- GET /forensics/evidence/{id}

22.2 Timeline API

- GET /forensics/timelines
- POST /forensics/timelines

22.3 Report API

- GET /forensics/reports
- POST /forensics/reports

22.4 Export API

- POST /forensics/export

23. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/  
■■■ src/  
■ ■■■ digital_forensics/  
■■■ tests/  
■ ■■■ digital_forensics/  
■■■ docs/  
■ ■■■ digital_forensics/  
■■■ api/  
■ ■■■ digital_forensics/  
■■■ archive/
```

No top-level repository modifications are permitted.

24. Security Architecture

The AQELYN Digital Forensics Engine is a trusted subsystem responsible for preserving the integrity, authenticity, and admissibility of digital evidence.

Every forensic operation shall be:

- Evidence-backed
- Cryptographically verified
- Policy-governed
- Auditable
- Traceable
- Reproducible
- Role-authorized
- Legally defensible

24.1 Security Principles

- Zero Trust
- Least Privilege
- Immutable Evidence
- Chain of Custody
- Defense in Depth
- Secure by Design
- Continuous Verification
- Separation of Duties

24.2 Authorization Model

Supported operational roles:

- Forensic Analyst
- Senior Forensic Analyst
- Incident Commander
- Evidence Custodian
- SOC Analyst
- Compliance Officer
- Security Administrator
- Automation Service

All privileged forensic operations shall be authorized through the AQELYN Policy Engine.

24.3 Forensic Integrity

Forensic records shall maintain:

- Unique identifier
- Evidence hash
- Collection timestamp
- Collector identity
- Custody state
- Verification state
- Analysis history
- Report references
- Audit history

Evidence history shall be append-only.

24.4 Chain-of-Custody Protection

Chain-of-custody records shall support:

- Immutable transfer records
- Actor attribution
- Time-stamped custody events
- Verification records
- Export records

Access records
Audit trail

No custody entry shall be destructively modified.

25. Digital Forensics Lifecycle

25.1 Evidence Lifecycle

Collected
↓
Verified
↓
Registered
↓
Analyzed
↓
Referenced
↓
Archived

25.2 Chain of Custody Lifecycle

Created
↓
Transferred
↓
Verified
↓
Audited
↓
Closed

25.3 Investigation Lifecycle

Opened
↓
Artifact Collection
↓
Analysis
↓
Timeline Reconstruction
↓
Findings
↓
Report Generated
↓
Archived

25.4 Report Lifecycle

Draft
↓
Reviewed
↓
Approved
↓
Published
↓
Archived

26. Continuous Verification

The engine continuously validates:

- Evidence integrity
- Hash consistency
- Chain of custody
- Artifact availability
- Storage integrity
- Repository consistency

27. Performance Requirements

The engine shall support:

- Large forensic datasets
- Parallel artifact analysis
- High-speed hash verification
- Scalable timeline reconstruction
- Concurrent investigations
- Efficient report generation

28. Scalability Requirements

The engine shall scale to support:

- Millions of forensic artifacts
- Large disk images
- Large memory images
- Enterprise investigations
- Distributed storage
- Hybrid deployments

29. Audit Requirements

Every forensic operation shall generate immutable audit records.

Audit events include:

- Evidence acquisition
- Evidence verification
- Custody transfer
- Artifact analysis
- Timeline generation
- Report publication
- Evidence export

30. Failure Handling

30.1 Evidence Acquisition Failure

- Acquisition halted
- Failure recorded
- Retry permitted

30.2 Verification Failure

- Integrity failure recorded
- Evidence quarantined
- Investigator notified

30.3 Storage Failure

Evidence preserved
Recovery initiated
Audit generated

30.4 Export Failure

Export cancelled
Integrity maintained
Audit logged

31. Testing Strategy

31.1 Unit Testing

Validate:

Acquisition Manager
Verification Service
Timeline Engine
Memory Analysis
Disk Analysis
Log Analysis
Report Generator

31.2 Integration Testing

Verify interaction with:

Kernel
Evidence Engine
Knowledge Graph
Event Bus
SOC Engine
Risk Intelligence
Threat Intelligence
Workflow Engine
Policy Engine

31.3 System Testing

Validate:

Evidence acquisition
Timeline reconstruction
Artifact analysis
Hash verification
Report generation
Evidence export

31.4 Security Testing

Verify:

Authorization
Integrity validation
Chain of custody
Audit logging
Policy enforcement

31.5 Regression Testing

Verify IS-001 through IS-015 remain unaffected.

32. Acceptance Criteria

IS-016 is complete when:

```
Evidence acquisition implemented
Chain of custody implemented
Integrity verification implemented
Timeline engine implemented
Artifact analysis implemented
Report generation implemented
Repository unchanged
Testing documented
```

33. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/digital_forensics/
■■■ tests/digital_forensics/
■■■ docs/digital_forensics/
■■■ api/digital_forensics/
■■■ archive/
```

No top-level repository modifications are permitted.

34. Engineering Summary

IS-016 introduces the AQELYN Digital Forensics Engine, providing enterprise-grade forensic acquisition, preservation, analysis, reporting, and chain-of-custody management.

Major capabilities include:

```
Evidence Acquisition
Evidence Verification
Chain of Custody
Artifact Repository
Timeline Reconstruction
Memory Analysis
Disk Analysis
Log Analysis
Registry Analysis
Browser Artifact Analysis
Forensic Reporting
Evidence Export
```

The engine integrates with the Evidence Engine, Knowledge Graph, Security Operations Engine, Risk Intelligence Engine, and Compliance Engine while maintaining repository stability, evidence integrity, and backward compatibility.

35. Specification Status

```
Specification ID : IS-016
Title : AQELYN Digital Forensics Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0016
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
```



```
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
EA-0011 COMPLETE
EA-0012 COMPLETE
EA-0013 COMPLETE
EA-0014 COMPLETE
EA-0015 COMPLETE
IS-016 COMPLETE
EA-0016 READY FOR GENERATION
```

36. EA-0016 Engineering Objective

The objective of IS-016 was to introduce a dedicated Digital Forensics Engine that enables AQELYN to acquire, preserve, verify, analyze, report, and export digital evidence with full chain of custody and forensic integrity.

The engine extends AQELYN from security operations into forensic defensibility and reproducible investigations.

37. EA-0016 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Forensic acquisition
Chain of custody
Evidence verification
Artifact repository management
Timeline reconstruction
Memory analysis
Disk analysis
Log analysis
Registry analysis
Browser artifact analysis
Forensic reporting
Evidence export
Knowledge Graph integration
Evidence Engine integration
Security Operations support
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

38. Major Engineering Decisions

38.1 Decision 1 - Dedicated Digital Forensics Engine

Forensic acquisition, preservation, and analysis responsibilities are implemented as a standalone engine rather than embedded in the Evidence Engine or SOC Engine.

Rationale:

```
Clear separation of evidence storage from forensic examination.
Independent lifecycle and scaling.
Better support for chain-of-custody and forensic reporting.
Improved legal and regulatory defensibility.
```

38.2 Decision 2 - Evidence Engine Remains Source of Truth

The Digital Forensics Engine consumes and extends evidence managed by the AQELYN Evidence Engine but does not replace it.

Benefits:

- Preserves evidence architecture.
- Avoids duplication of authoritative evidence records.
- Allows forensic artifacts to reference immutable evidence.
- Maintains consistency with existing evidence workflows.

38.3 Decision 3 - Chain of Custody as First-Class Object

Chain of custody is modeled as a Universal Object Model extension.

Benefits:

- Custody becomes queryable.
- Evidence transfer history becomes auditable.
- Legal defensibility improves.
- Knowledge Graph can link custody to evidence, actors, and investigations.

38.4 Decision 4 - Event-Driven Forensics

Evidence acquisition, verification, analysis, timeline generation, report generation, and export events are published through the AQELYN Event Bus.

Examples include:

- evidence.acquired
- evidence.verified
- evidence.exported
- analysis.started
- analysis.completed
- timeline.generated
- report.generated
- report.exported

This maintains loose coupling between AQELYN engines.

38.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

- EvidenceArtifact
- ChainOfCustody
- Timeline
- ForensicReport

These extend the Universal Object Model without modifying existing object definitions.

39. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | EvidenceArtifact, ChainOfCustody, Timeline, ForensicReport objects |

| IS-003 Event Bus | Evidence, analysis, timeline, report, and export events |

| IS-004 Evidence Engine | Immutable evidence records, artifact metadata, hash validation |

| IS-005 Knowledge Graph | Evidence, artifact, timeline, investigation, incident, hash relationships |

| IS-006 Trust Engine | Evidence trust and confidence |

| IS-007 Mission Engine | Mission impact context for forensic reports |

| IS-008 Workflow Engine | Forensic review, approval, export, and custody workflows |

| IS-009 Policy Engine | Evidence retention, access, custody, export, destruction policies |

| IS-010 Compliance Engine | Audit evidence, compliance investigations, control validation |

| IS-013 Risk Intelligence Engine | Evidence supporting risk and incident impact validation |

| IS-014 Threat Intelligence Engine | Threat artifacts and investigation context |

| IS-015 SOC Engine | Incident evidence, timeline analysis, forensic reports |

No existing engine required redesign.

40. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/digital_forensics/  
■■■ tests/digital_forensics/  
■■■ api/digital_forensics/  
■■■ docs/digital_forensics/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

41. Security Impact Summary

The specification introduces digital-forensics-specific security controls:

```
Cryptographic evidence verification  
Immutable chain of custody  
Role-authorized forensic operations  
Policy-governed evidence access  
Evidence export authorization  
Forensic audit trail  
Evidence quarantine on verification failure  
Reproducible forensic reporting
```

No reduction in the security posture of existing components was identified.

42. Capabilities Added

The engine enables AQELYN to support:

```
Digital evidence acquisition  
Evidence preservation  
Chain of custody management  
Artifact indexing  
Evidence hashing  
Evidence verification  
Timeline reconstruction  
Host forensics  
File system analysis  
Memory analysis
```

Registry analysis
Browser artifact analysis
Log correlation
Forensic reporting
Evidence export

43. Risks Identified

| Risk | Mitigation |

|---|---|

Evidence integrity failure	Hash verification and quarantine
Chain-of-custody gaps	Immutable custody records and transfer logging
Unauthorized evidence export	Policy enforcement and role authorization
Large evidence volume	Scalable repository and parallel analysis
Analysis reproducibility issues	Timeline and report provenance
Storage failure	Recovery workflow and audit generation
Evidence access ambiguity	Access logging and actor attribution
Report publication errors	Review and approval lifecycle

No critical architectural risks were identified that require redesign.

44. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover evidence acquisition, chain of custody, integrity verification, timeline engine, artifact analysis, report generation, repository validation, and testing documentation.

45. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

46. Dependencies

Required:

EA-0001 through EA-0015
IS-001 through IS-015

Enables:

IS-017 and subsequent detection, analytics, investigation, and forensic-dependent components

47. Completion Record

Engineering Archive : EA-0016
Implementation Specification : IS-016
Title : AQELYN Digital Forensics Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

48. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine

49. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0016

Current Status:
EA-0016 COMPLETE

Next Implementation Specification:
IS-017 - AQELYN Threat Detection & Analytics Engine

EA-0016 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-017.

50. Engineering Archive Publication Standard

EA-0016 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
├──
├── diagrams/
│   ├── Architecture.svg
│   ├── Component.svg
│   ├── Workflow.svg
│   ├── EventFlow.svg
│   └── Integration.svg
├──
├── examples/
│   └── example_*.md
├──
├── HTML/
│   ├── index.html
│   ├── styles.css
│   └── assets/
├──
├── images/
├──
├── journal/
│   └── Engineering_Journal.md
├──
├── manifest/
│   └── manifest.json
├──
├── MD/
│   └── EA-xxxx.md
├──
├── PDF/
│   └── EA-xxxx.pdf
├──
├── requirements/
│   └── Requirements_Matrix.md
├──
├── traceability/
│   └── Traceability_Matrix.md
├──
└── README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

51. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-016-001	Support evidence acquisition	Sections 8, 12	Complete
FR-016-002	Maintain chain of custody	Sections 8, 12, 25	Complete
FR-016-003	Verify evidence integrity	Sections 8, 12, 24	Complete
FR-016-004	Reconstruct timelines	Sections 8, 12, 25	Complete
FR-016-005	Analyze forensic artifacts	Sections 8, 12	Complete
FR-016-006	Generate forensic reports	Sections 8, 12, 25	Complete
FR-016-007	Publish forensic events	Sections 8, 15, 38	Complete
NFR-016-001	Immutable evidence handling	Sections 9, 24, 41	Complete
NFR-016-002	Cryptographic verification	Sections 9, 12, 24	Complete
NFR-016-003	Full auditability	Sections 9, 29, 41	Complete
NFR-016-004	Evidence traceability	Sections 9, 14, 39	Complete
NFR-016-005	Scalable storage integration	Sections 9, 27, 28	Complete
NFR-016-006	Repository stability	Sections 23, 33, 40	Complete

52. Traceability Matrix

| Source | Target | Relationship |

|---|---|---|

| IS-016 Purpose | EA-0016 Objective | Defines why the engine exists |

| Acquisition Manager | FR-016-001 | Implements forensic acquisition |

| Chain of Custody Manager | FR-016-002 | Implements custody tracking |

| Evidence Verification Service | FR-016-003 | Implements hash validation |

| Timeline Engine | FR-016-004 | Implements timeline reconstruction |

| Artifact Analysis Engines | FR-016-005 | Implements artifact analysis |

| Report Generator | FR-016-006 | Implements forensic reporting |

| Event Publisher | FR-016-007 | Publishes forensic events |

| Evidence Engine Integration | Evidence source of truth | References immutable evidence |

| Knowledge Graph Integration | Evidence relationships | Links artifacts, timelines, investigations, incidents |

| SOC Integration | IS-015 | Supports incident investigation |

| Risk Integration | IS-013 | Provides evidence supporting risk |

| Compliance Integration | IS-010 | Supports audit and regulatory reporting |

| Policy Integration | Security rules | Controls retention, access, custody, export |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

53. Engineering Journal

Journal Entry - EA-0016

EA-0016 was created to archive completion of IS-016 - AQELYN Digital Forensics Engine.

The archive records the expansion of AQELYN into digital forensics. IS-016 defines the structure needed to acquire evidence, verify integrity, maintain chain of custody, index artifacts, reconstruct timelines, analyze memory, disk, logs, registry, browser artifacts, generate forensic reports, and export evidence packages.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Digital forensics must be modeled separately from evidence storage and SOC operations. The Evidence Engine remains the source of truth for evidence records, the SOC Engine consumes forensic output for investigations, and the Digital Forensics Engine owns acquisition, analysis, custody, verification, timeline reconstruction, and forensic reporting.

Governance Note

EA-0016 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

54. Examples

54.1 Example Evidence Artifact

```
artifact_id: ART-0001
evidence_id: EVID-0001
artifact_type: memory_image
source: endpoint-001
hash:
sha256: 7a0f...example
collected_at: 2026-07-07T12:00:00Z
```

54.2 Example Chain of Custody

```
custody_id: COC-1001
evidence: EVID-0001
collector: forensic_analyst_01
timestamp: 2026-07-07T12:05:00Z
transfers:
- from: forensic_analyst_01
to: evidence_custodian_01
timestamp: 2026-07-07T12:30:00Z
verified: true
```

54.3 Example Timeline

```
timeline_id: TL-2001
events:
- timestamp: 2026-07-07T11:55:00Z
event: suspicious_process_started
evidence: EVID-0001
- timestamp: 2026-07-07T12:01:00Z
event: outbound_connection_detected
evidence: EVID-0002
```

54.4 Example Forensic Event

```
{
  "event_type": "evidence.verified",
  "evidence_id": "EVID-0001",
  "hash_algorithm": "SHA-256",
  "verification_status": "passed",
  "source_engine": "aqelyn_digital_forensics_engine"
}
```

55. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0016.md
PDF/EA-0016.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_forensics.md
```

56. Final Archive Statement

EA-0016 is the Engineering Archive for IS-016 - AQELYN Digital Forensics Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

EA-0016 COMPLETE
IS-016 COMPLETE
NEXT: IS-017